學士學位 論文

On the Origin of Quantum Discreteness in Plasma Electron Scattering

: A Non-Archimedean Exploration of the Amplituhedron and Its p-adic Structure

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Undergraduate Thesis

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1 Summary of the thesis

1.1 The limit that does not vanish

In the semiconductor etching process, below the 2 nm node, a single etch trench holds only tens to hundreds of ions. At this scale, statistical averages lose their physical meaning.

Particle scarcity – a 2 nm trench holds only tens to hundreds of ions and 6 to 7 Si atoms.
Particle trajectory scattering – ions deviate from their estimated straight paths in the sheath.
Radical and Ion placement uncertainty – radicals adsorb randomly onto the surface, and the collisions that chemical reaction requires become random in place and time.

Line edge roughness does not scale down with the feature. As devices shrink, its relative impact grows.

1.2 The question

Monte Carlo, surrogate models, LLM-based predictors – every method that describes this regime speaks in the language of probability. However, this thesis steps one question behind them.

Is the probabilistic character of the microscopic world a property of nature? Or an artifact of the real-number language chosen to describe it?

1.3 Where the question becomes concrete

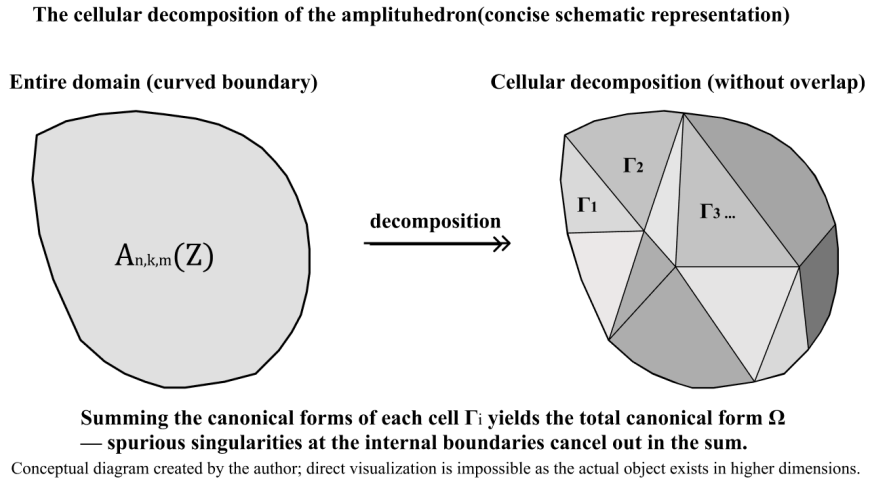


FIGURE 1.1: The cellular decomposition of the amplituhedron (concise schematic representation)

The amplituhedron is a geometric object whose canonical form yields a scattering amplitude – with no reference to Feynman diagrams, probability, or spacetime.

A setting where the hypothesis can be tested by computation, not argument.

1.4 How it was computed

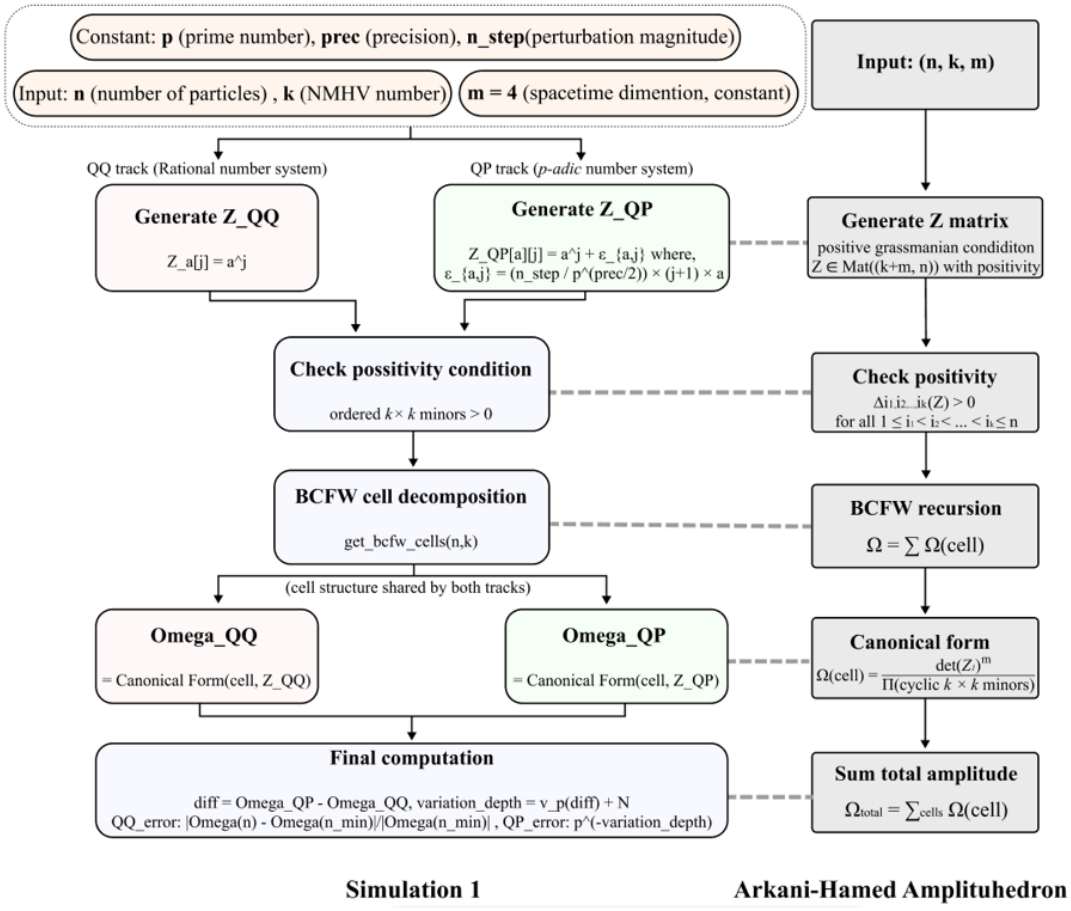


FIGURE 1.2: Flowchart of the simulation algorithm and its correspondence with Arkani-Hamed & Trnka, *The Amplituhedron*, JHEP 10 (2014) 030.

The canonical form Ω is evaluated along two parallel tracks – QQ over the rationals, QP under a p-adic perturbation of the same input. Both in exact rational arithmetic. Implemented in Python, verified against five independent invariants.

1.5 What did not happen

The hypothesis predicted a gain in precision. It did not appear. Both tracks are exact and there is no error to improve upon. **However**, the hypothesis was not rejected. It was redirected – from magnitude to valuation.

1.6 What did happen

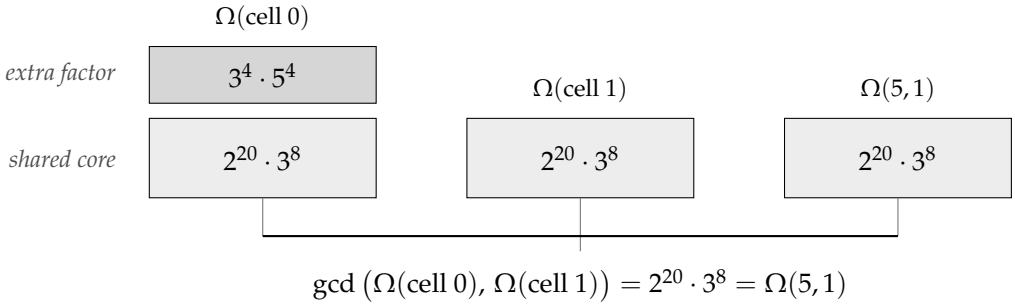


FIGURE 1.3: The shared arithmetic core of the two BCFW cells of $\Omega(7, 2)$. Both cells carry the common core $2^{20} \cdot 3^8$; cell 0 differs only by the extra factor $3^4 \cdot 5^4$. The greatest common divisor is therefore the core itself, which equals the lower amplitude $\Omega(5, 1)$—the arithmetic content of $\Omega(5, 1)$ surviving as a divisor of both cells of the higher amplitude.

Measured by p-adic valuation instead of magnitude, the two tracks differ by a definite and reproducible arithmetic structure. Both BCFW cells of $\Omega(7, 2)$ carry the common core $2^{20} \cdot 3^8$ – which is exactly $\Omega(5, 1)$, the lower amplitude. Nothing was designed to produce this. It is deposited by the moment-curve construction and propagated by the recursion.

1.7 Where it points

The result is narrow: the choice of number system leaves a definite, reproducible, and measurable imprint on an exactly computed amplitude. The forward computation cannot go further. The two number systems are expected to part in the inverse problem – recovering, from a given amplitude, the input that produced it.

Over the **reals**, the candidate inputs form a continuum.
Over the p**-adics**, they resolve into a discrete set.

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